

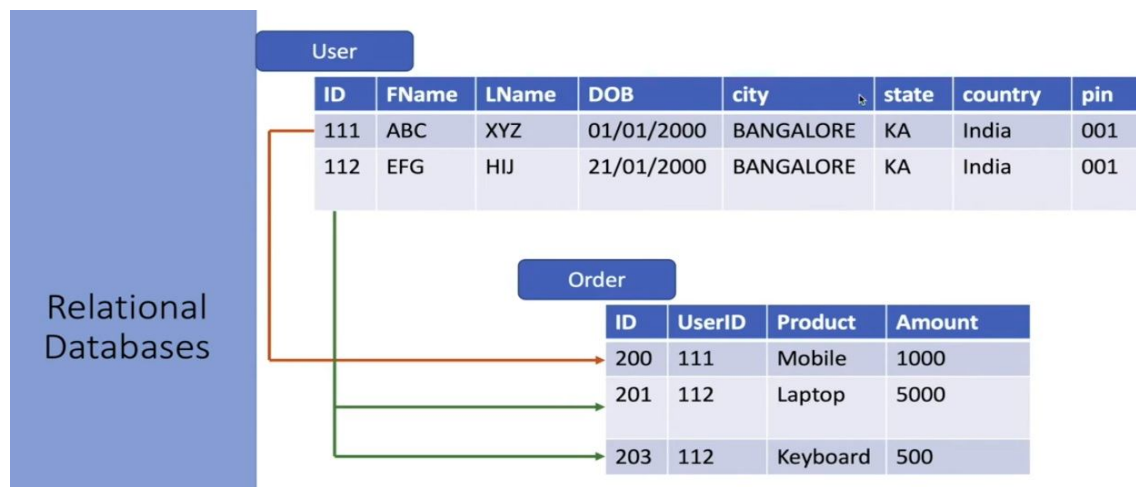
What is NoSQL. Explain its types.?

NoSQL database means “Not Only SQL”, Non-Relational, Non tabular database and Schema less database.

Comparison with SQL and NoSQL.

Feature	SQL	NoSQL
Schema	Fixed	Flexible
Scalability	Vertical	Horizontal
Data type	Structured	Semi / Unstructured
Transactions	ACID	BASE (Basically Available, Soft State, Eventually Consistent)

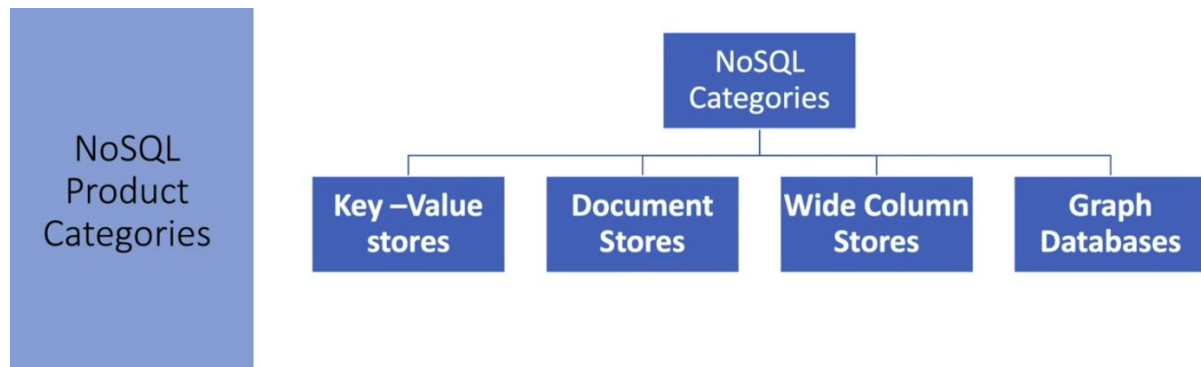
Relational Database



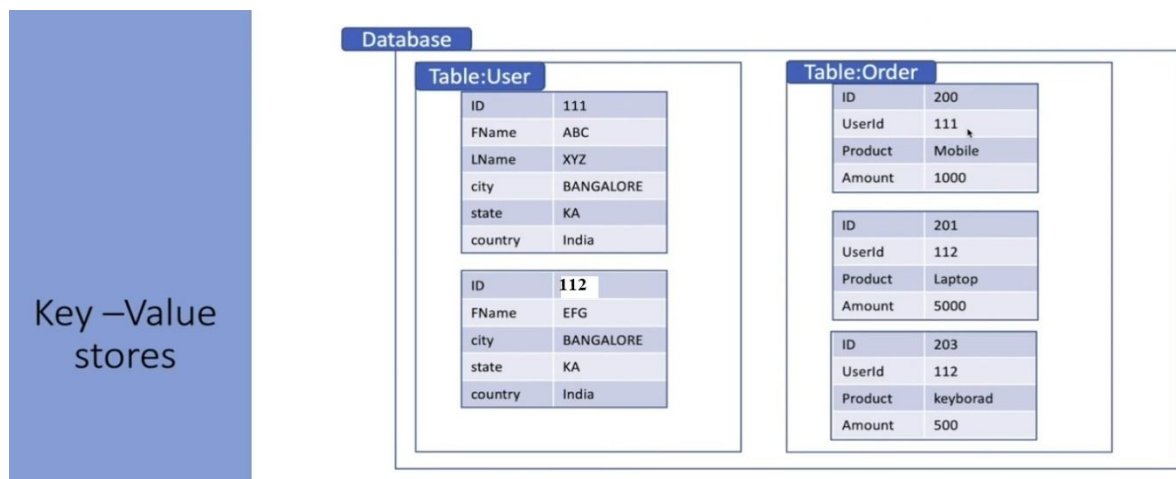
NOSQL



Types of NoSQL Databases



1. Key Value Stores: Data is stored as **key-value pairs**, Very fast and simple, Best for Caching, Session Data



Example structure:

Key → Value

ID → 111

Examples:

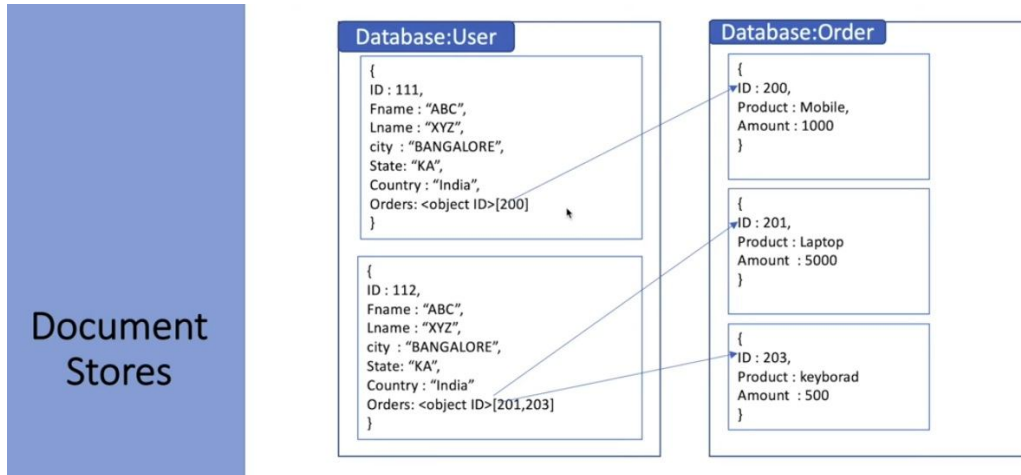
- Redis
- DynamoDB
- Riak

Use cases:

- Caching
- User sessions
- Shopping carts

2. Document-Oriented Database

- Data stored as **documents** (JSON, BSON, XML)
- Flexible schema
- Each document can have different fields



Example:

```
{
  "ID": 111,
  "FName": "ABC",
  "LName": "XYZ",
  "City": "BANGALORE",
  "State": "KA",
  "Country": "India",
  "Order": ["201", "203"]
}
```

Examples:

- MongoDB
- CouchDB
- Firestore

Use cases:

- Content management
- Web applications
- APIs

Comparison Between Key Value Store and Document Store

Feature	Key-Value Store	Document Store
Data Format	Key → Value (simple)	Key → JSON Document
Structure Awareness	No	Yes
Query Capability	By key only	By key + fields
Indexing	Usually key only	Multiple fields
Complexity	Very simple	More flexible
Example	Redis	MongoDB

3. Wide Column Store

- Data stored in **columns instead of rows**,
- Optimized for large datasets and analytics

Wide Column Stores

Table:User		
ID	111	
Super Column : Name		
Column : FName	ABC	
Column : LName	XYZ	
Super Column : Address		
Column : city	BANGALORE	
Column : state	KA	
Column : country	India	
Super Column : Orders		
Orders	[200, 201]	

ID	111	
Super Column : Name		
Column : FName	ABC	
Column : LName	XYZ	

Table:Order		
ID	200	
Super Column : PRODUCT		
Column : Product	Laptop	
Super Column : Price		
Column : Amount	5000	

ID	201	
Super Column : PRODUCT		
Column : Product	Keyboard	
Super Column : Price		
Column : Amount	500	

Examples:

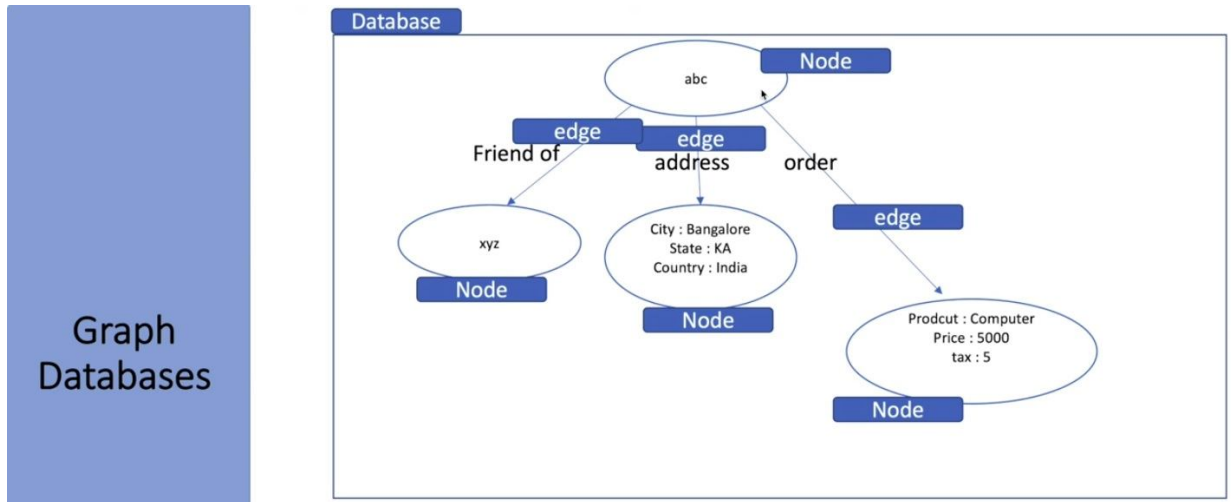
- Apache Cassandra
- HBase
- Google Bigtable

Use cases:

- Big data processing
- Time-series data
- Data warehousing

4. Graph Database:

- Data stored as **nodes and edges**
- Focuses on relationships



Examples:

- Neo4j
- Amazon Neptune
- ArangoDB

Use cases:

- Social networks
- Recommendation systems
- Fraud detection